Context-Scoped AI Agents in Integrated Developer Workspaces: A Between-Subjects Study of Information Breadth and Task Performance

ANDREW SADLER, Michigan Technological University, USA
JIE JW WU, Michigan Technological University, USA

Software development is fragmented across issue trackers, team chat, and code hosts, and AI assistants embedded in that workflow face an analogous fragmentation: they can only act on what they are allowed to see. We operationalise this as a *context ladder*—a three-rung experimental manipulation of how much of a shared workspace an AI agent can observe and act on: chat only (C_1), chat and task board (C_2), or chat, board, and GitHub (C_3)—and measure the downstream effect on user task performance, cognitive load, and trust across a population spanning both novice and experienced programmers.

We present Weld, an open-source integrated developer workspace (SvelteKit + Rust/Axum) combining a Kanban board, team chat, and mocked GitHub repository data in one product, and report a hybrid user study (target $N = 20$–30) recruiting from two complementary populations: university CS students (undergraduate and graduate, representing less-experienced developers) and professional software developers ($\geq$1 year of industry experience). Every participant first completes two manual no-agent baseline tasks— establishing a per-person speed and workload anchor that absorbs individual skill differences—then is randomly assigned to one agent condition for six agent-assisted tasks spanning comprehension, single-surface, and cross-surface coordination types. Including both populations allows us to examine whether the benefit of context breadth varies with developer experience, and whether novice and experienced programmers differ in their reliance on and trust of an AI agent with varying information access. Primary analysis uses baseline-normalised task time via a mixed-effects model testing the condition × task-type interaction; secondary measures include NASA-TLX, System Usability Scale, a five-item integration-perception scale, and an eight-item AI trust scale.

Context scoping is enforced at the server in code—not in prompts—by the Context Gate module, which records a per-turn `agent_context_sent` audit snapshot proving, from the log alone, exactly what information each agent invocation received. This architectural choice makes the experimental manipulation verifiable rather than merely assumed. We contribute the context-ladder study design, the mixed-experience participant protocol, and the Context Gate mechanism as replicable instruments for empirical research on AI agents in multi-surface collaborative tools.

CCS Concepts: • **Software and its engineering** → **Software development techniques**; • **Human-centered computing** → **HCI theory, concepts and models**; **Empirical studies in HCI**; • **Computing methodologies** → *Artificial intelligence*.

Additional Key Words and Phrases: AI agents, developer tools, integrated workspaces, context scoping, user study, NASA TLX, cognitive load, task coordination, human-AI interaction, software engineering

Authors' addresses: Andrew Sadler, amsadler@mtu.edu, Michigan Technological University, Houghton, Michigan, USA; Jie JW Wu, jie.jw.wu@gmail.com, Michigan Technological University, Houghton, Michigan, USA.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.
Manuscript submitted to ACM

ACM Reference Format:
Andrew Sadler and Jie JW Wu. 2026. Context-Scoped AI Agents in Integrated Developer Workspaces: A Between-Subjects Study of Information Breadth and Task Performance. *ACM Trans. Softw. Eng. Methodol.* 1, 1 (June 2026), 14 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 INTRODUCTION

Modern software development demands constant movement across tools: a task management board to plan work, a chat platform to coordinate with teammates, and a code-hosting service to review pull requests and track issues. Each transition carries a context-switching cost—the cognitive overhead of re-establishing mental state in a new tool—that fragments attention and increases workload [4, 12]. This fragmentation is compounded when information lives in separate silos: acting on a chat message that references a board card requires the developer to open two applications and mentally merge their contents. Ko et al. [8] documented this cross-tool information-seeking overhead in collocated software teams; LaToza et al. [9] found it a primary driver of knowledge loss and coordination breakdowns.

Integrated developer workspaces—products that co-locate boards, chat, and code review—attempt to reduce this overhead by design. Yet the empirical record on whether integration alone suffices, and what further benefit is conferred by an embedded AI agent that can *observe and act* across all surfaces, remains thin. Prior AI-productivity research has focused on single-surface tasks (code generation, text drafting), leaving the cross-surface coordination layer largely unstudied [3, 13]. Agentic frameworks such as ReAct [15] and Reflexion [14] show strong performance on synthetic benchmarks but have not been evaluated with human participants on realistic multi-surface workflows.

A second open question concerns *context breadth*: how much workspace state should an agent be allowed to observe? Wider context may improve agent relevance but raises concerns about auditability and trust. Existing systems gate context via prompt instructions, which are opaque and unverifiable—there is no way to audit, after the fact, exactly what information an agent received during a specific turn.

This paper makes three contributions:

(1) **Weld system.** We describe Weld, an open-source all-in-one developer workspace (Kanban board, team chat, GitHub integration) with an embedded study mode that provisions isolated per-participant workspaces, implements balanced condition assignment, and logs fine-grained behavioral telemetry (§3).
(2) **Code-enforced context scoping (Context Gate).** We introduce the Context Gate, a server-side module that assembles agent context from a condition-specific database allowlist, with a per-turn audit snapshot (`agent_context_sent`) that proves—from the server log alone— exactly which information sources a given agent turn could access (§3).
(3) **Empirical study.** We report a controlled between-subjects user study ($N = TBD$) crossing three agent-context conditions (C_1 chat-only advisory; C_2 chat+board action; C_3 chat+board+GitHub full) against three task types (comprehension, single-surface, cross-surface), measuring task completion, time-on-task, NASA-TLX, SUS, integration perception, AI trust, and cross-surface behavioral telemetry (§5).

The remainder of the paper is structured as follows. Section 2 surveys related work on tool fragmentation, AI productivity agents, and context management. Section 3 describes Weld and its study infrastructure. Section 4 presents the study design. Section 5 reports results. Section 6 discusses implications, and Section 7 concludes.

2 RELATED WORK

2.1 Fragmentation and Context-Switching in Software Development

The cognitive cost of task interruption and tool-switching in knowledge work has been well-documented. Mark et al. [12] showed through in-situ observation that interrupted workers worked faster to compensate but reported higher stress and effort. Czerwinski et al. [4] found through diary studies that workers switched tasks on average every three minutes, with complex task resumption taking substantially longer than simple ones. In software engineering specifically, Ko et al. [8] identified information fragmentation across tools as a primary bottleneck for collocated teams: developers spent significant time locating and reconciling information scattered across email, wikis, issue trackers, and chat. LaToza et al. [9] found that maintaining accurate mental models across multiple tools was a persistent challenge, with developers frequently reverting to conversations to reconstruct shared state. Together, these findings motivate the core premise of integrated developer workspaces: that reducing inter-tool navigation distance reduces cognitive load and improves task throughput.

Commercial responses—Linear (boards + notifications), Notion (docs + tasks), GitHub Projects (issues + boards), Slack with app integrations—have proliferated, yet the degree to which they reduce the measured costs identified by the above studies has not been evaluated in controlled experiments with realistic multi-surface task batteries. We address this gap in our baseline (no-agent) phase, which directly measures cross-surface task performance in an integrated environment.

2.2 AI Agents for Developer Productivity

The introduction of large language model (LLM) assistants into developer workflows has produced a growing body of empirical evidence. Chen et al. [3] demonstrated that LLMs could generate functionally correct code for a substantial fraction of benchmark problems, and GitHub Copilot [6] subsequently reported that developers accepted a significant fraction of AI-generated suggestions in production use. Noy and Zhang [13] showed in a randomized controlled trial that access to an LLM writing assistant improved task completion time and output quality for college-educated workers. These studies focus on single-surface, largely *generative* tasks—writing code, text, or structured content.

The evaluation of AI agents in *coordination* tasks—where the agent must synthesize information across surfaces and take multi-step actions—is less developed. Work on agentic LLM frameworks such as ReAct [15] and Reflexion [14] demonstrates strong performance on synthetic benchmark tasks (web navigation, knowledge-base QA, code execution), but these benchmarks do not capture the human-in-the-loop, multi-surface collaboration dynamics present in real developer work. Studies of LLM agents in CSCW contexts—where an agent participates in a team workflow rather than assisting a single user on a single surface—are largely absent from the literature.

WELD's embedded agent operates in this understudied space: it can read the team's chat history, inspect and modify the Kanban board, and link cards to GitHub artifacts—all within a single turn. Crucially, the agent's actions are not auto-applied; board mutations and drafted chat messages are surfaced to the participant for review, keeping the human in the loop for consequential actions. This design reflects recent findings [13] that humans are effective editors of AI-generated outputs even when they cannot independently generate them.

2.3 Context Management in LLM-Based Systems

A central design challenge for LLM-powered tools is *context management*: deciding what information to include in the model's context window and at what granularity. In retrieval-augmented generation (RAG) systems [10], this is handled by a retrieval step that selects relevant passages from a corpus. In multi-agent systems, each agent typically receives a curated subset of shared state. The trade-off between context richness and noise (or cost, in token-priced systems) has been analysed from an accuracy standpoint but rarely from the perspective of user trust, cognitive load, or task-type interaction.

Prior deployed systems typically gate context in one of two ways. The first approach is *prompt-level* gating: the system prompt instructs the model to ignore or not request certain information sources. This is easy to implement but is neither verifiable nor robust—the model may still reference disallowed information if it was included upstream, and there is no audit trail proving that a given agent turn operated within intended bounds. The second approach is *integration-level* gating: only the API calls corresponding to allowed sources are made before the model is invoked. This is the approach taken by WELD's CONTEXT GATE.

The auditability of context decisions is closely related to the broader theme of transparency in AI systems. Dey's foundational framework [5] defines context as "any information that can be used to characterise the situation of an entity," and argues that context-aware systems must make their information selection criteria explicit to maintain user trust. Cavoukian's privacy-by-design principles [2] extend this to governance: systems should embed privacy and data minimisation into their architecture, not bolt it on. The CONTEXT GATE is an instance of this principle applied to agent information access: what the agent cannot see by construction, it cannot leak, misuse, or be manipulated into retrieving.

2.4 User Studies of AI-Augmented Tools

Studies of AI-augmented developer tools have primarily measured productivity via proxy (task completion rate, time, code quality). Fewer studies employ validated psychometric instruments alongside behavioral measures. Our study combines NASA-TLX [7]—a widely used multidimensional workload scale—with the System Usability Scale [1] and a trust scale adapted from Madsen and Gregor [11], enabling comparison with prior HCI work on tool usability and AI trust. The pairing of subjective self-report scales with objective behavioral telemetry (time-on-task, cross-surface event counts) allows triangulation that neither source alone affords: participants may report low workload on tasks where telemetry shows long completion times, or vice versa, and this discrepancy is itself informative.

3 SYSTEM: WELD

3.1 Architecture Overview

Weld is an open-source, full-stack developer workspace targeting teams of two to ten engineers. The web frontend is built with SvelteKit 2 (TypeScript, Tailwind CSS 4); the backend is an Axum 0.8 server (Rust) backed by PostgreSQL 16 and Redis. A WebSocket hub broadcasts real-time board and chat updates to all connected clients of a team. All study-mode paths share the same production application binary, activated by an `appConfig.studyMode` flag.

Weld exposes three collaboration surfaces that correspond exactly to the three context sources manipulated across the agent conditions:

Kanban Board. Columns and cards with fractional-indexed positions, real-time drag-and-drop reordering, label and priority assignment, and GitHub issue/PR links. Cards carry a `github_issue` and `github_pr` field that are populated either by the user or by the agent's `link_card_to_github` tool. Cross-references to chat messages render inline as `[[card:uuid]]` tokens that resolve to card titles on hover.

Team Chat. Channel-based messaging with threading, Markdown rendering, @mention resolution, and the same `[[card:uuid]]` cross-reference syntax used on the board. Messages are delivered over WebSocket and persisted server-side; the agent's `post_message` tool drafts a message that the participant must review and confirm before it is sent.

GitHub Integration. A GitHub App installation synchronizes open issues and pull requests into Weld's read-only GitHub view. Two-way linking (card ↔ issue, card ↔ PR) is supported both manually and via the agent's `link_card_to_github` tool.

3.2 Study Mode and Workspace Isolation

Each study participant is provisioned in a single atomic database transaction via the `provision_participant` function. The transaction creates: a dedicated user account, a team with four pre-seeded synthetic teammates (Sarah Chen, Alex Rivera, Priya Nair, and Marcus Webb), a PetTracker project, three board columns (Backlog, In Progress, Done), 15–20 cards with realistic descriptions and priorities, two channels (#general and #pettracker) with pre-seeded message history, and GitHub fixture data. All timestamps in the seeded content are stored as negative offsets from the provisioning moment, so the workspace always appears to have been active for the same relative period regardless of when the participant joins.

Access codes and resumption. Participants authenticate via single-use access codes rather than email/-password registration. If a participant closes the browser mid-session, they can return to the same code on any device and resume from their saved stage and task index (`resume_stage`, `resume_task_index`). Progress is persisted by `update_progress(pool, user_id, stage, task_index)`.

Condition assignment. The `pick_agent_condition` function implements balanced permuted-block randomization with block size 3 and a random tie-break on equal counts, ensuring that no condition falls behind the others as enrollment proceeds while avoiding deterministic C1→C2→C3 cycling that could confound condition with enrollment order.

Database isolation. Every query in study mode is scoped by `team_id` and `project_id` in the SQL `WHERE` clause. No cross-participant query paths exist; the study database schema enforces this at the row level.

3.3 The Context Gate

The Context Gate is a Rust module (`services/study/agent/gate.rs`) that acts as the sole chokepoint for assembling the agent's context before each LLM call. Its design centers on two structures: `ContextRules`, which encodes the per-condition allowlist, and `ScopedContext`, the assembled result.

Rules and allowlists. `rules_for(condition)` returns a `ContextRules` value specifying which of the three sources (chat, board, GitHub) may be queried. The rules for each condition are:

Condition	Chat	Board	GitHub
C_1 (advisory)	✓	–	–
C_2 (action)	✓	✓	–
C_3 (full)	✓	✓	✓

The gate's `assemble(pool, ctx)` function queries *only* the allowed sources; the database calls for disallowed sources are never made. The resulting `ScopedContext` carries a `sources_included` field listing exactly which sources were consulted, a `chat: Vec<ChatLine>` field, an `board: Option<Vec<CardLine>>`, and a `github: Option<Vec<GithubItem>>`. Disallowed fields are `None`.

Proof-of-scoping snapshot. Before invoking the LLM, `run_turn` calls `ScopedContext::snapshot()`, which serialises the assembled context (minus any fields that should not appear in logs) and records it as an `agent_context_sent` telemetry event. This creates an immutable per-turn audit record proving that the context assembly respected the assigned condition.

Double-gated tool executor. The gate also governs which tools are available to the model. `allowed_tool_categories(condition)` returns a static slice of `ToolCategory` variants: C_1 receives `[]`, C_2 receives `[BoardWrite, ChatPost]`, and C_3 receives all three categories (`[BoardWrite, ChatPost, GithubLink]`). This list determines which tool definitions are sent to the model in the OpenAI request (first gate). The tool executor (`tools::execute`) independently re-validates the tool name against the condition's allowed categories before executing any action; any call that slips past the first gate is rejected and logged as `agent_tool_rejected` (second gate). Table 1 lists the five tools and their condition availability.

Table 1. Agent tools by condition availability. ✓ = available, – = not offered to model and rejected at executor.

Tool name	Action	C_1	C_2	C_3
`create_card`	Create a board card	–	✓	✓
`move_card`	Move a card to another column	–	✓	✓
`update_card`	Edit card priority / assignee / description	–	✓	✓
`post_message`	Draft a chat message for review	–	✓	✓
`link_card_to_github`	Link a card to an issue or PR	–	–	✓

Turn budget. Each agent invocation is bounded by two hard limits: `MAX_TURNS = 5` model↔tool round-trips per invocation, and `MAX_AGENT_MESSAGES = 50` total agent messages per participant across the entire session. These limits prevent runaway loops and bound infrastructure cost.



Fig. 1. The CONTEXT GATE architecture. The `assemble()` call queries only allowed sources for the participant's assigned condition and emits a `agent_context_sent` snapshot. The double-gated tool executor re-validates tool permissions independently at execution time, logging violations as `agent_tool_rejected` events.

3.4 Agent Turn Orchestration

The `run_turn` function orchestrates each agent invocation. After assembling the `ScopedContext` and recording the `agent_context_sent` event, it loads prior turns from the `study_agent_turns` table to maintain conversation continuity across the session. The function then enters a loop bounded by `MAX_TURNS`, alternating between LLM calls (with condition-appropriate tool definitions from `openai_tool_defs(condition)`) and tool execution calls.

Condition-specific system prompt. The system prompt varies by condition. For C_1, it instructs the model to "provide guidance but not suggest or execute actions," constraining it to an advisory role even though no tools are offered. For C_2 and C_3, the prompt frames the agent as "a co-worker in the team workspace" that can take actions on behalf of the participant. This framing difference is intentional: it matches participant expectations to the agent's actual capability.

Chat message drafting. The `post_message` tool does not immediately post a message; instead, the tool executor returns a `ChatDraft` to the client, which renders it as a review panel. The participant must explicitly confirm before the message is sent. This keeps human oversight over public communication while still reducing the mechanical effort of composing and locating the correct channel.

3.5 Telemetry Infrastructure

WELD records 43 discrete event types (Table 2). Client events are batched and submitted to a single endpoint that validates each event type against a server-side allowlist (`ALLOWED_EVENT_TYPES`) before persistence; events not on the allowlist are silently dropped. This *allowlist-by-construction* approach means that even

a modified client cannot introduce new event types, preventing accidental capture of keystrokes, scroll positions, or other data outside the study protocol.

Server-side events (agent lifecycle, context snapshots, tool outcomes) are recorded via `record_server_event` directly within the Axum handlers, bypassing the client batch path and therefore unaffected by client connectivity or manipulation.

Table 2. Telemetry event categories and types recorded by Weld.

Category	Event types
Session & phase	`session_start`, `session_end`, `session_abandoned`, `session_resumed`, `phase_started`, `phase_ended`, `consent_given`
Survey	`survey_started`, `survey_completed`
Task lifecycle	`task_shown`, `task_completed`, `task_skipped`, `task_final_state`, `task_rating`, `instruction_reopened`
Board	`card_created`, `card_opened`, `card_edited`, `card_moved`, `card_assigned`, `card_labeled`, `card_priority_set`, `card_deleted`, `card_searched`, `board_filtered`
Chat	`message_sent`, `message_edited`, `message_deleted`, `channel_switched`, `thread_opened`, `mention_clicked`, `link_clicked`
Cross-surface	`card_referenced_from_chat`, `chat_referenced_from_card`, `pr_referenced_from_card`, `card_referenced_from_pr`, `issue_referenced_from_card`
GitHub	`github_view_opened`, `pr_opened`, `issue_opened`, `pr_linked_to_card`, `issue_linked_to_card`
Navigation	`view_changed`, `search_used`
Agent	`agent_invoked`, `agent_context_sent`, `agent_response`, `agent_tool_call`, `agent_tool_rejected`, `agent_suggestion_accepted`, `agent_suggestion_rejected`, `agent_action_undone`, `agent_output_edited`, `agent_error`, `agent_panel_opened`, `agent_panel_collapsed`

4 STUDY DESIGN

4.1 Research Questions

RQ1 *Integrated workspace value (baseline).* Does using an integrated workspace (without an agent) reduce cognitive load and time-on-task relative to reported experience with fragmented tooling?

RQ2 *Effect of context breadth on agent effectiveness.* Does increasing the agent's context breadth ($C_1 \rightarrow C_2 \rightarrow C_3$) improve task performance and reduce cognitive load?

RQ3 *Task-type moderation.* Does the effect of context breadth vary across task types (comprehension, single-surface, cross-surface)?

RQ4 *Scaling of benefit with surface integration.* Do participants in higher-context conditions show greater cross-surface integration behavior (measured by telemetry) and higher integration perception scores?

4.2 Hypotheses

Based on the research questions and prior work on AI-assisted workflows, we state four directional hypotheses:

H1 (Context breadth and speed). Broader agent context reduces time-on-task: participants in C_3 will complete agent tasks faster than those in C_2, who will in turn be faster than those in C_1.

H2 (Task-type moderation). The effect of context breadth is moderated by task type. The advantage of C_3 over C_1 will be negligible for comprehension and single-surface tasks (where cross-surface data is irrelevant) but substantial for cross-surface tasks that require coordinating board, chat, and GitHub.

H3 (Cognitive load). Broader context reduces perceived cognitive load (NASA-TLX): $C_3 < C_2 < C_1$, driven primarily by effort and mental-demand subscales.

H4 (Perception–behavior gap). Trust and integration-perception scores will not fully track objective task performance: participants may report high trust or high integration perception even in conditions where telemetry shows limited cross-surface coordination, and vice versa.

4.3 Participants

We target $N = 20$–30 participants drawn from two complementary populations: *university CS students* (undergraduate and graduate, recruited via course mailing lists) and *professional software developers* ($\geq$1 year of industry experience, recruited via professional networks and online communities). Recruiting both groups allows us to examine whether the benefit of context breadth varies with prior experience. Eligibility further required self-reported familiarity with at least one task-management tool (e.g., Jira, Linear, GitHub Projects) and one team-chat tool (e.g., Slack, Teams, Discord). The study was conducted remotely. Participants were compensated *[amount TBD]*.

4.4 Conditions

Participants were randomly assigned, with balanced block randomization (block size 3), to one of three between-subjects conditions:

C_1 **(Chat only, advisory).** The agent observes only team chat history. No access to board state or GitHub. Advisory-only: no tools offered to the model, so it cannot modify any surface.

C_2 **(Chat + Board, action-capable).** The agent observes chat and the full board card state. Available tools: `create_card`, `move_card`, `update_card`, and `post_message` (drafts a chat message for the participant to review before sending).

C_3 **(Chat + Board + GitHub, full action).** The agent observes all three surfaces. All C_2 tools plus `link_card_to_github`, enabling it to connect board cards to GitHub issues or pull requests.

4.5 Tasks

The study uses a hybrid design. Each participant completes 8 tasks in fixed order: two manual *baseline tasks* (C0, no agent available) establishing a per-person speed and workload anchor, followed by six *agent-assisted tasks* in their assigned condition. Baseline and agent tasks are content-disjoint to prevent learning transfer. The agent phase uses the `gpt-4o-mini-2024-07-18` model at temperature 0.2 with versioned condition-specific system prompts pinned for reproducibility.

Task types span both phases:

Comprehension. Synthesize information from one or more surfaces to form a judgment (e.g., "what does the team need most right now?").

Single-surface. Can be completed by acting on a single surface, even if cross-surface context helps.

Cross-surface. Explicitly requires coordinated action across two or more surfaces.

Table 3. Task design summary. B = Baseline (no agent, C0), T = Agent task. Surfaces: Ch = Chat, Bo = Board, GH = GitHub.

ID	Phase	Type	Surfaces	Summary
B1	Baseline	Single-surface	Ch, Bo	Find blocked card from chat; move to In Progress
B2	Baseline	Cross-surface	Ch, Bo	Check card status; post reply in chat
T1	Agent	Comprehension	Ch, Bo	Determine next priority; update board
T2	Agent	Comprehension	Ch, Bo, GH	Catch-up summary; post in chat
T3	Agent	Single-surface	Ch	Draft and post status update in chat
T4	Agent	Single-surface	Bo	Set priorities on three backlog cards
T5	Agent	Cross-surface	Ch, Bo	Find bug card from chat; move to In Progress
T6	Agent	Cross-surface	Ch, Bo, GH	Link PR to card; announce in chat

4.6 Procedure

Each session proceeded as follows:

(1) Informed consent and eligibility screening.
(2) Orientation to Weld (un-timed, guided).
(3) **Baseline phase** (2 tasks, no agent, condition C0): tasks presented sequentially; participants self-paced, rated each task before advancing.
(4) Phase-transition interstitial: agent panel revealed; C_2/C_3 participants received a brief explanation of available agent actions.
(5) **Agent phase** (6 tasks): same structure with the agent available under the participant's assigned condition (C_1, C_2, or C_3).
(6) Exit survey (NASA-TLX, SUS, integration-perception, AI trust scale, open-ended comments, demographics).
(7) Debrief.

Sessions were conducted remotely. Screen recording was *not* collected; all behavioral data came from in-product telemetry.

4.7 Measures

Task performance. Task completion (binary, judged from `task_final_state` telemetry) and time-on-task (seconds between `task_shown` and `task_completed` or `task_skipped`).

Cognitive load. NASA-TLX [7], six subscales (mental demand, physical demand, temporal demand, performance, effort, frustration), administered once per phase in the exit survey.

Usability. SUS [1], 10 items, 5-point Likert, yielding a 0–100 score.

Integration perception. A five-item scale measuring whether participants perceived workspace surfaces as connected (e.g., "I could easily see how chat discussions connected to board tasks").

AI trust and effectiveness. An eight-item scale (adapted from [11]) measuring perceived agent accuracy, helpfulness, and appropriate scope.

Behavioral telemetry. Cross-surface reference events (`card_referenced_from_chat`, `pr_linked_to_card`, etc.) as an objective measure of integration behavior; agent turn counts and tool-call distributions as agent-use indicators.

5 RESULTS

5.1 Participant Demographics and Data Quality

5.2 RQ1: Integrated Workspace Value (Baseline)

Table 4. Baseline-phase NASA-TLX subscale means (SD) by condition. [PLACEHOLDER]

Subscale	C_1	C_2	C_3
Mental Demand	—	—	—
Temporal Demand	—	—	—
Effort	—	—	—
Frustration	—	—	—
Overall Weighted	—	—	—

5.3 RQ2: Effect of Context Breadth

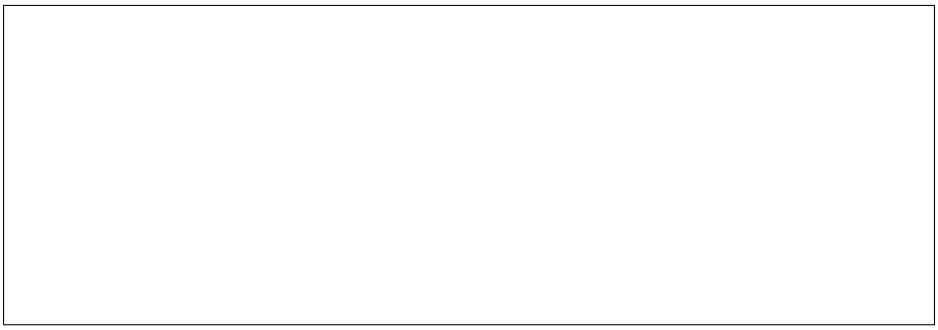


Fig. 2. Task completion time (seconds) by condition and task type. Error bars show 95% CI. [PLACEHOLDER]

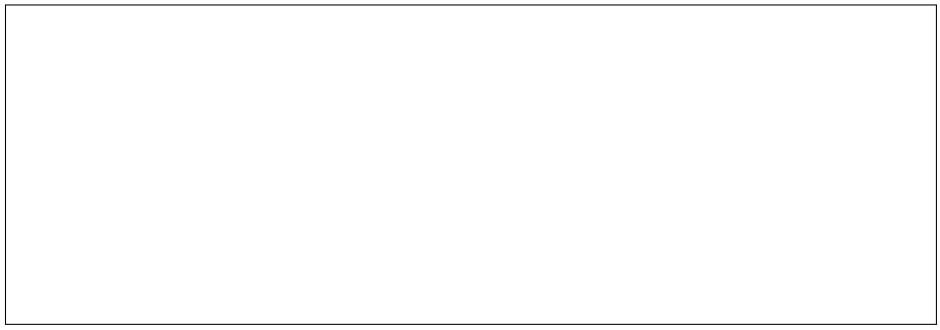


Fig. 3. NASA-TLX score by condition and task type (comprehension / single-surface / cross-surface). [PLACEHOLDER]

Table 5. Cross-surface telemetry event counts and integration-perception scale means (SD) by condition. [PLACEHOLDER]

Measure	C_1	C_2	C_3
Cross-surface events (median)	—	—	—
Integration perception (mean)	—	—	—
AI trust scale (mean)	—	—	—

5.4 RQ3: Task-Type Moderation

5.5 RQ4: Integration Behavior and Perception

6 DISCUSSION

6.1 Context Breadth as a Design Variable

6.2 Task Type as a Moderator

6.3 Code-Enforced Context Scoping as a Design Principle

The Context Gate pattern—enforcing context boundaries in server code rather than model prompts—has implications beyond this study. First, it makes the manipulation mechanically verifiable: any reviewer can inspect the `agent_context_sent` audit logs and confirm that a C_1 agent never received board data. Second, it suggests a governance model for production AI agents in collaborative tools, where organizations need auditable controls over what information agents can observe. Third, it decouples the *scope* of an agent from its *capability*, allowing researchers and practitioners to vary context breadth independently of model capability.

6.4 Limitations

7 CONCLUSION

We presented Weld, an integrated developer workspace with a built-in study mode, and a between-subjects user study ($N = TBD$) that isolates the effect of agent context breadth on task performance, cognitive load, and cross-surface coordination behavior. The study's primary methodological contribution is the Context Gate: a server-side, code-enforced context-scoping mechanism that makes the agent's information diet both

configurable and auditable, enabling a clean between-subjects manipulation without relying on prompt engineering.

Our results [to be reported] suggest [summary of key empirical findings]. Together, these findings have implications for the design of AI agents in collaborative productivity tools, particularly regarding (1) the value of gradually expanding context rather than granting full information access by default, and (2) the importance of audit-logging context assembly decisions as a trust and governance mechanism.

Future work should examine longitudinal effects of integrated AI agents on team coordination patterns, study generalization to professional teams working on real projects, and explore adaptive context scoping—where the agent's information diet expands or contracts based on task characteristics detected at runtime.

Weld and the study infrastructure are open-source and available at https://github.com/weldtool/weld.

ACKNOWLEDGMENTS

The authors thank the study participants for their time. [Additional acknowledgements to be added before submission.]

REFERENCES

[1] John Brooke. 1996. SUS: A "Quick and Dirty" Usability Scale. In *Usability Evaluation in Industry*. Taylor & Francis, 189–194.

[2] Ann Cavoukian. 2009. *Privacy by Design: The 7 Foundational Principles*. Technical Report. Information and Privacy Commissioner of Ontario.

[3] Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, et al. 2021. Evaluating Large Language Models Trained on Code. *arXiv preprint arXiv:2107.03374* (2021).

[4] Mary Czerwinski, Eric Horvitz, and Susan Wilhite. 2004. A Diary Study of Task Switching and Interruptions. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM, 175–182. https://doi.org/10.1145/985692.985715

[5] Anind K. Dey. 2001. Understanding and Using Context. *Personal and Ubiquitous Computing* 5, 1 (2001), 4–7. https://doi.org/10.1007/s007790170019

[6] GitHub. 2022. *Research: Quantifying GitHub Copilot's Impact on Developer Productivity and Happiness*. Technical Report. GitHub, Inc.

[7] Sandra G. Hart and Lowell E. Staveland. 1988. Development of NASA-TLX (Task Load Index): Results of Empirical and Theoretical Research. In *Advances in Psychology*. Vol. 52. Elsevier, 139–183. https://doi.org/10.1016/S0166-4115(08)62386-9

[8] Andrew J. Ko, Robert DeLine, and Gina Venolia. 2007. Information Needs in Collocated Software Development Teams. In *Proceedings of the 29th International Conference on Software Engineering (ICSE)*. IEEE, 344–354. https://doi.org/10.1109/ICSE.2007.45

[9] Thomas D. LaToza, Gina Venolia, and Robert DeLine. 2006. Maintaining Mental Models: A Study of Developer Work Habits. In *Proceedings of the 28th International Conference on Software Engineering (ICSE)*. ACM, 492–501. https://doi.org/10.1145/1134285.1134355

[10] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, et al. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33. 9459–9474.

[11] Melissa Madsen and Shirley Gregor. 2000. Measuring Human-Computer Trust. In *Proceedings of the 11th Australasian Conference on Information Systems*. 6–8.

[12] Gloria Mark, Daniela Gudith, and Ulrich Klocke. 2008. The Cost of Interrupted Work: More Speed and Stress. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM, 107–110. https://doi.org/10.1145/1357054.1357072

[13] Shakked Noy and Whitney Zhang. 2023. Experimental Evidence on the Productivity Effects of Generative Artificial Intelligence. *Science* 381, 6654 (2023), 187–192. https://doi.org/10.1126/science.adh2586

[14] Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: Language Agents with Verbal Reinforcement Learning. In *Advances in Neural Information Processing*

Systems (NeurIPS), Vol. 36.

[15] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *International Conference on Learning Representations (ICLR)*.